

Electromagnetics — In-Semester Exam Quick Revision

PYQ-focused • Electrostatics • Magnetostatics • Stokes Theorem

How to use: First memorize the boxed formulas, then practice the PYQ methods. In numerical problems, always write the governing law before substituting values.

1. Electrostatics — Core Formulas

Topic	Formula / Key Result
Coulomb's law	$F = (1 / 4\pi\epsilon_0) \cdot q_1q_2 / r^2$
Electric field	$E = F/q$; for point charge: $E = (1 / 4\pi\epsilon_0) \cdot q/r^2$
Electric flux density	$D = \epsilon E$
Gauss law	$\oint E \cdot dS = Q_{\text{enc}}/\epsilon_0$
Potential	$V = (1 / 4\pi\epsilon_0) \cdot q/r$
Field–potential relation	$E = -\nabla V$
Dipole moment	$p = qd$
Dipole torque	$\tau = pE \sin\theta$
Energy density	$u = \frac{1}{2} \epsilon E^2$
Capacitance	$C = Q/V$; parallel plate: $C = \epsilon A/d$
Stored energy	$W = \frac{1}{2} CV^2$

2. Poisson & Laplace Equations

Poisson: $\nabla^2 V = -\rho_v/\epsilon_0$

Laplace: $\nabla^2 V = 0$ (charge-free region)

Remember: If charge density exists $\rightarrow$ Poisson. If $\rho_v = 0 \rightarrow$ Laplace.

3. Boundary Conditions

Electric field: $E_{1t} = E_{2t}$

Electric flux density: $D_{2n} - D_{1n} = \rho_s$

Magnetic flux density: $B_{1n} = B_{2n}$

Magnetic field: $H_{1t} - H_{2t} = J_s$ (sign depends on chosen normal)

4. Magnetostatics — Core Formulas

Law / Quantity	Formula / Key Result
Biot–Savart law	$dB = (\mu_0/4\pi) \cdot I(dl \times R)/R^2$
Ampère's circuital law	$\oint H \cdot dl = I_{\text{enc}}$
Magnetic flux density	$B = \mu H$
Infinite straight wire	$B = \mu_0 I / (2\pi r)$
Current density	$J = \nabla \times A$
Vector magnetic potential	$B = \nabla \times A$
Scalar magnetic potential	$H = -\nabla V_m$ (in current-free region)
Lorentz force	$F = q(E + v \times B)$

5. Faraday's Law

$$\blacksquare \oint \mathbf{E} \cdot d\mathbf{l} = - \frac{d\Phi_B}{dt}$$

For N turns: $\text{emf} = -N \frac{d\Phi}{dt}$. The negative sign represents Lenz's law.

6. Stokes Theorem — VERY IMPORTANT

Statement: The line integral of a vector field around a closed contour equals the surface integral of its curl over the surface bounded by that contour.

$$\blacksquare \oint_C \mathbf{A} \cdot d\mathbf{l} = \blacksquare_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

Use: Convert a closed line integral into a surface integral, or vice versa. Orientation of C and dS must follow the right-hand rule.

PYQ-Oriented Numerical Methods

7. Potential $\rightarrow$ E $\rightarrow$ D $\rightarrow$ Charge Density

Typical question: Given $V(x,y,z)$, find V at a point, E , D and ρ_v .

1. Substitute the coordinates directly into V to obtain potential.
2. Find E using $E = -\nabla V$.
3. Find D using $D = \epsilon E$.
4. Find ρ_v using Poisson: $\rho_v = -\epsilon \nabla^2 V$.

Worked PYQ-style example

$$V = 3xyz^2 + 5xz - 3y^2z, \text{ at } A(3,7,4)$$

$$V(A) = 3(3)(7)(4^2) + 5(3)(4) - 3(7^2)(4) = 480 \text{ V}$$

$$\partial V / \partial x = 3yz^2 + 5z$$

$$\partial V / \partial y = 3xz^2 - 6yz$$

$$\partial V / \partial z = 6xyz + 5x - 3y^2$$

Therefore $E = -[(\partial V / \partial x)a_x + (\partial V / \partial y)a_y + (\partial V / \partial z)a_z]$. Substitute (3,7,4) only after differentiating.

8. Magnetic Field / Current Density Numerical

If H is given: $\mathbf{J} = \nabla \times \mathbf{H}$. Use the Cartesian curl determinant:

$$\nabla \times \mathbf{H} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ H_x & H_y & H_z \end{vmatrix}$$

Then calculate total current from $I = \oint \mathbf{J} \cdot d\mathbf{S}$. Write the surface limits carefully.

9. Coulomb Force on Multiple Charges

Use $F = (1/4\pi\epsilon_0) \cdot q_1 q_2 / r^2$ for each pair.

Draw the geometry and identify directions before adding forces.

Resolve each force into x - and y -components.

Add components: $F_x = \sum F_{x_i}$, $F_y = \sum F_{y_i}$.

Resultant: $F = \sqrt{(F_x^2 + F_y^2)}$, and $\theta = \tan^{-1}(F_y / F_x)$.

10. Ampère / Surface-Current Numerical

If $\mathbf{J}$ is given over a surface: $I = \oint \mathbf{J} \cdot d\mathbf{S}$.

For spherical coordinates, remember: $dS = r^2 \sin\theta \, d\theta \, d\phi$ for a spherical surface.

Then use Ampère's law: $\oint \mathbf{H} \cdot d\mathbf{l} = I_{enc}$.

11. Most Important Theory Questions to Prepare

1. Gauss's law — statement, derivation and applications.
2. Poisson and Laplace equations — derivation, difference and applications.
3. Faraday's law — integral/differential form and physical meaning.
4. Vector magnetic potential and relation $\mathbf{B} = \nabla \times \mathbf{A}$.
5. Scalar magnetic potential and its use.
6. Boundary conditions for E , D , B and H .
7. Lorentz force equation.
8. Stokes theorem — statement, mathematical form and application.
9. Biot–Savart law and Ampère's circuital law.

12. Last-Minute Exam Checklist

Before the exam	✓
Memorize Gauss, Ampère, Faraday and Stokes laws	■
Practice ∇V , $\nabla^2 V$ and $\nabla \times A/H$	■
Practice one Coulomb-force vector problem	■
Practice one $V \rightarrow E \rightarrow D \rightarrow \rho v$ problem	■
Practice one $J = \nabla \times H$ and $I = \oint J \cdot dS$ problem	■
Revise boundary conditions and Lorentz force	■

Exam tip: In every derivation, write the law first, define the symbols, show the main mathematical steps, and finish with the boxed final result. In numericals, include units and vector directions.